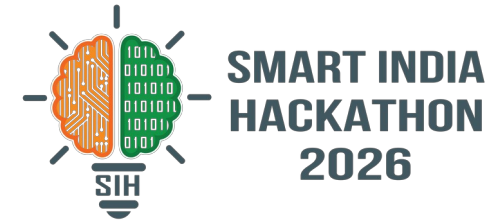
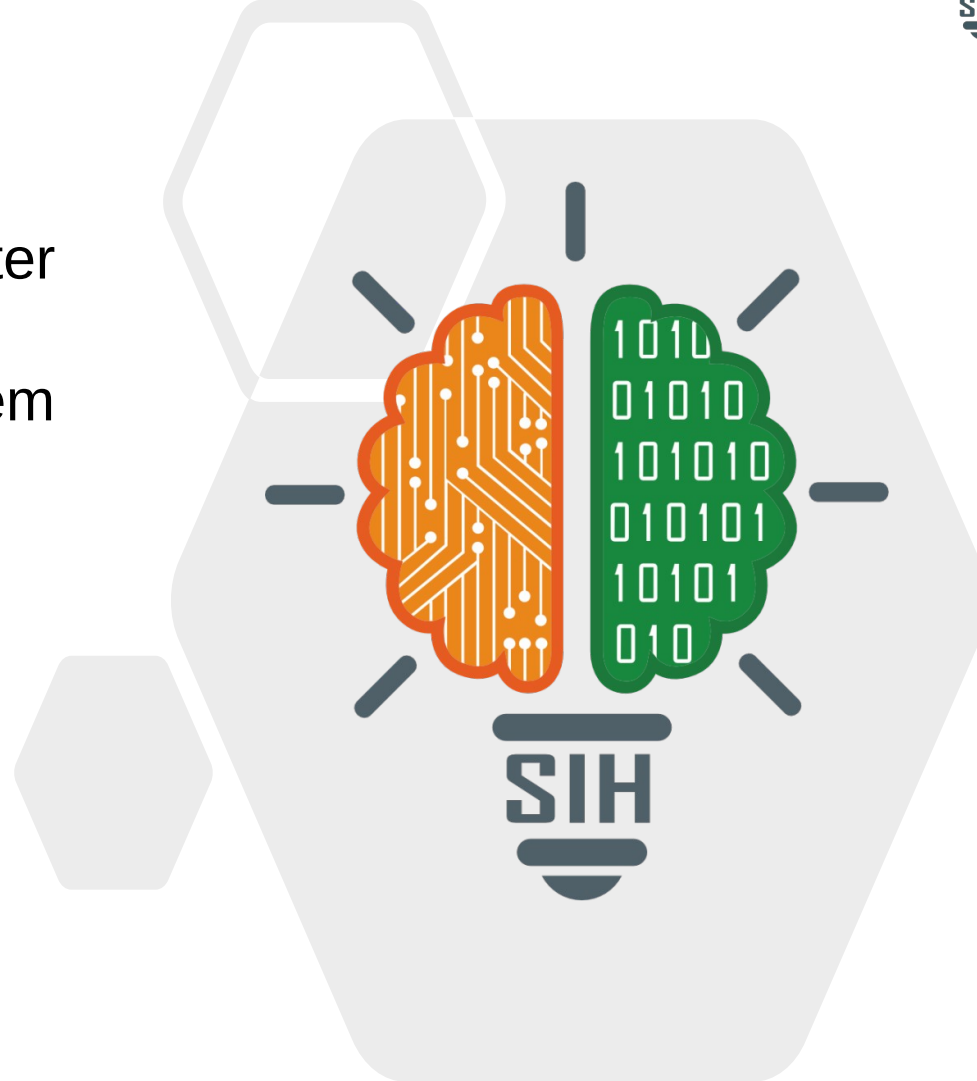


SMART INDIA HACKATHON 2026



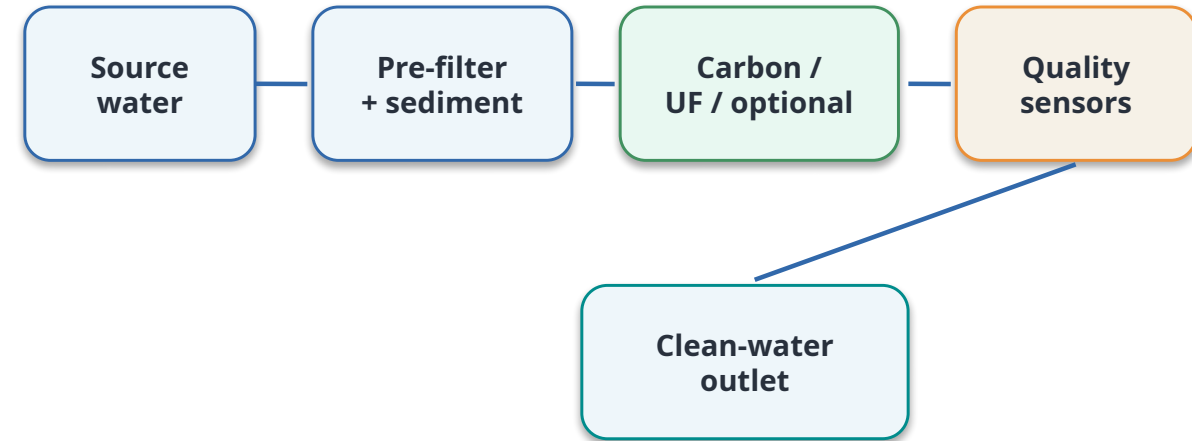
- **Problem Statement ID** – 26040
- **Problem Statement Title**- Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas
- **Theme**- Clean & Green Technology
- **PS Category**- Hardware
- **Team ID**- TX-SIH26-112
- **Team Name** – INFRA MINDS



Proposed Solution:-

- Solar-powered water treatment system designed for remote communities
- Continuously monitors pH, turbidity, TDS, and temperature to ensure safe water quality
- Provides automatic alerts/shut-off when limits are exceeded

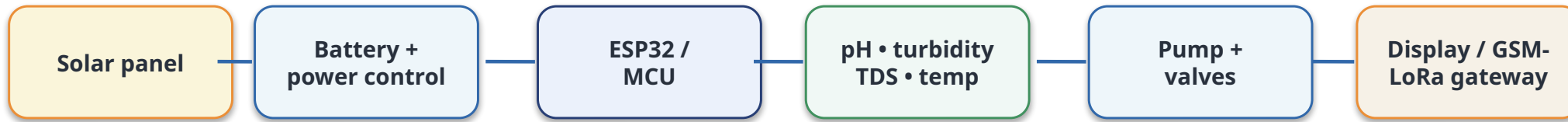
How the system works



Innovation: treatment + monitoring + energy autonomy in one modular platform, designed for easy deployment and maintenance where infrastructure is limited.

TECHNICAL APPROACH

System Architecture & Implementation Flow



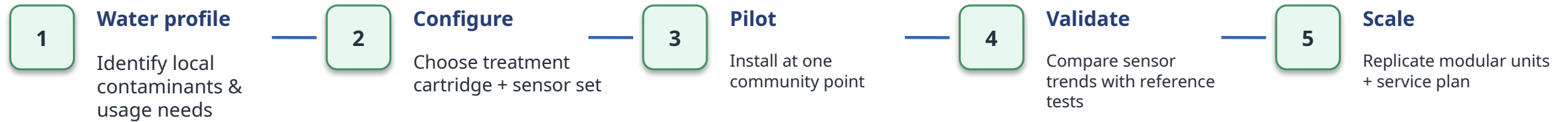
Treatment train



TECHNOLOGIES:

- 1.ESP32-class microcontroller, pH, turbidity, TDS & temperature sensors
- 2.DC pump + solenoid valves + Solar panel + battery + low-power control

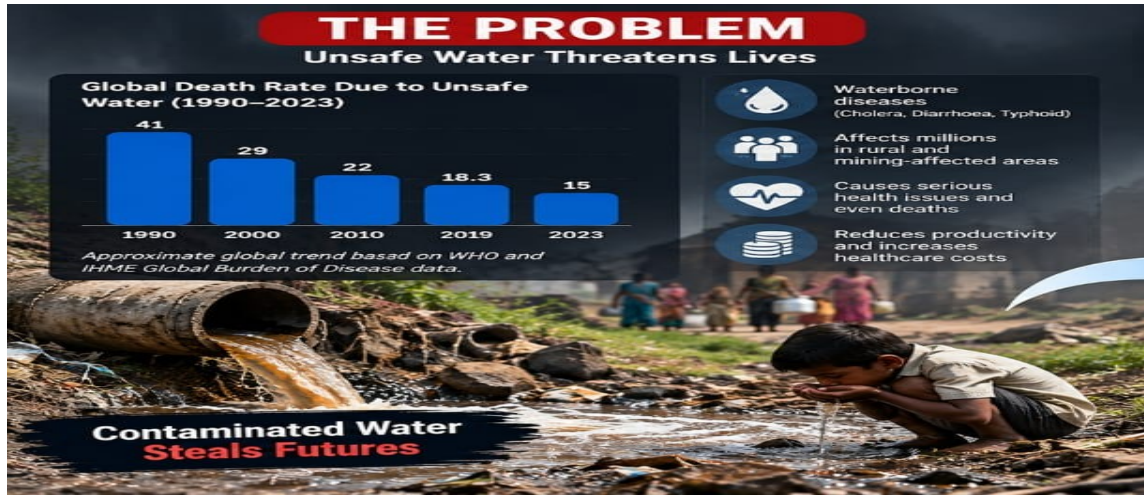
Practical deployment plan



Challenges & mitigations

Changing turbidity	Modular pre-filtration + replaceable cartridges
Mining-related contamination	Site-specific testing; validated adsorption/RO stage if required
Limited electricity	Solar + battery + low-power electronics
Sensor drift	Regular calibration + reference sample checks
Remote maintenance	Simple modules + local operator training



Mining-affected
areas

Earlier indication of changing quality conditions

Rural households

Accessible treatment + water-quality awareness

Environment

Solar operation + modular treatment footprint

Local operators

Status display, alerts and maintenance cues

Goal: make water-quality management proactive, measurable and easier to operate in remote settings.

REFERENCE BASE FOR THE PROPOSED CONCEPT

Grounded in trusted guidelines. Designed for real-world impact.

 **CLEAN WATER
HEALTHY PEOPLE
SUSTAINABLE COMMUNITIES****01 SIH 2026 – Problem Statement 26040**Government of Jharkhand –
Clean & Green Technology –
Hardware**02 WHO – Guidelines for drinking-water quality (2026)**Health-based targets,
risk management and
surveillance for
drinking-water safety.**03 WHO – Small water supplies guideline (2024)**Practical risk management
for small/community
drinking-water supplies.**04 Jal Jeevan Mission – WQ Monitoring & Surveillance Framework**Water-quality testing,
monitoring and
remedial action for rural
drinking-water supply.**05 Jal Jeevan Mission – WQ Training / WQMIS resources**Resources for field testing,
sample testing and
water-quality information
management.**FROM
GUIDELINES
TO GREATER
LIVES****CLEAN WATER
FOR RURAL AND
MINING-AFFECTED
COMMUNITIES**

BASED ON EVIDENCE. DRIVEN BY PEOPLE. FOR A HEALTHIER TOMORROW.